

Phase-by-Phase Execution Plan: Production-Grade RAG System

This document defines a strict, time-boxed execution plan for building a production-grade Retrieval-Augmented Generation (RAG) system within 4–5 days, assuming ~8 focused hours per day. This is an engineering-first plan, not a tutorial.

Day 1 – System Design & Data Foundations

- 1 • Finalize system scope and non-goals (no chat UI, no prompt experimentation).
- 2 • Design end-to-end architecture diagram (ingestion, storage, retrieval, generation, API).
- 3 • Define metadata schema: `document_id`, `version`, `section`, `effective_date`, `department`.
- 4 • Select vector DB and relational DB and define responsibility split.
- 5 • Create repository structure and service boundaries.

Day 2 – Document Ingestion & Chunking

- 1 • Implement document loaders (PDF, DOCX, Markdown).
- 2 • Parse documents into structured sections.
- 3 • Implement section-aware chunking logic (no fixed-size splitting).
- 4 • Attach metadata and versioning to every chunk.
- 5 • Store metadata in relational DB.

Day 3 – Embeddings & Retrieval Pipeline

- 1 • Implement embedding service abstraction.
- 2 • Store embeddings in vector database.
- 3 • Implement semantic similarity search.
- 4 • Add metadata filtering and top-k selection.
- 5 • Implement simple re-ranking logic.

Day 4 – Answer Generation & Guardrails

- 1 • Implement LLM adapter layer.
- 2 • Construct grounded prompt using retrieved chunks only.
- 3 • Add citation generation per answer section.
- 4 • Implement refusal logic for low-confidence retrieval.
- 5 • Detect conflicting sources and surface explicitly.

Day 5 – API, Deployment & Validation

- 1 • Build FastAPI endpoints (/ingest, /ask, /health).
- 2 • Add structured logging and trace IDs.
- 3 • Dockerize all services.
- 4 • Run end-to-end validation with test documents.
- 5 • Write README explaining design decisions and trade-offs.

Completion Criteria: The project is considered complete only if answers are grounded, citations are correct, the system refuses when data is insufficient, and the entire stack runs via Docker without notebooks.